

# Rahul Panwar

✉ panwarrahul3183@gmail.com    LinkedIn    GitHub    Portfolio   ☎ 7607480583

*AI Engineer building production RAG systems, LLM evaluation pipelines, and hybrid retrieval architectures with hands-on experience in document intelligence, structured extraction, and grounding strategies for high-stakes domains.*

## Experience

---

### AI Engineer | Karma AI

April 2026 – Present

*Built production RAG systems for document intelligence, structured extraction, and LLM evaluation across government agencies.*

- Built a RAG extraction pipeline converting unstructured case PDFs into structured JSON (witnesses, evidence, timeline, accused), extracting **~320 structured records from 55 pages** of test dossiers. (FastAPI · Qdrant · Celery · PostgreSQL)
- Engineered hybrid retrieval over **40k+ court judgments** using statutory section pre-filtering with semantic search reranking, reducing irrelevant results by **~97%** vs. similarity-only search. (Qdrant · Embeddings · Python)
- Achieved **~91% grounding rate** using anchor-span extraction, output whitelisting, and deterministic parsing replacing LLM calls, reducing hallucination in document intelligence pipelines. (NLP · Pydantic · Prompt Engineering)
- Built provider-agnostic LLM gateway via **LiteLLM**, swapping providers requires one config change with **zero re-indexing** of existing vectors. (LiteLLM · Azure OpenAI · REST API)
- Built LLM evaluation infrastructure: corruption/mutation testing (**87% defect detection, 0% false alarms**), independent LLM-as-judge with grounding checks, and golden regression suite. (LLM Evaluation · Python)

### Software Developer | FirstLease

Aug 2025 – April 2026

*Built an internal document intelligence system enabling natural-language search over client requirements for sales and operations teams.*

- Developed an internal RAG system using **LangChain**, **ChromaDB**, and an open-source LLM with custom document parsers, enabling natural-language Q&A over multi-page client requirement documents and cutting document-review time by an estimated **60%**. (LangChain · ChromaDB · Python)
- Built automated document ingestion pipeline handling PDFs, spreadsheets, and scanned documents with chunking, metadata extraction, and vector indexing for semantic retrieval across **500+ internal documents**. (Python · FastAPI · Docker)
- Integrated role-based access controls and conversation history tracking, making the system production-ready for **3 internal teams** across sales, operations, and compliance. (PostgreSQL · REST API · React)

## Education

---

Vellore Institute of Technology, B.Tech in Computer Science — AI & Robotics

2021 – 2025

## Technical Skills

---

- **AI & ML:** Large Language Models, RAG Pipeline Design, Vector Databases, Semantic Search, Prompt Engineering, LLM Evaluation, Hybrid Retrieval, Hallucination Mitigation, Agentic Workflows
- **Frameworks & Tools:** LangChain, LangGraph, Hugging Face Transformers, FastAPI, Flask, Celery, Redis, Docker, Ollama, LiteLLM, Pydantic, Node.js, React.js
- **Databases:** PostgreSQL, MongoDB, Qdrant, ChromaDB, FAISS
- **Cloud & APIs:** AWS (EC2), Azure OpenAI, OpenAI API, LiteLLM Gateway, REST APIs, Docker Compose
- **Languages:** Python, C++ , SQL